

Application of uniform experimental design theory to multi-strategy improved sparrow search algorithm for UAV path planning

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ABSTRACT

The sparrow search algorithm (SSA) is a meta-heuristic optimization algorithm based on the predatory behavior of sparrows. However, SSA tends to fall into the local optimum when solving optimization problems with complex constraints. To improve its optimization efficiency and overall performance, this paper develops a multi-strategy improved SSA (ISSA) based on uniform experimental design theory. Specifically, the wrap-around L_2 -discrepancy (WD), as a uniformity metric for uniform design of experiments, is fully utilized with macro-regulation, adaptive dynamic management strategy, and boundary redistribution management mechanism to quantify the population uniformity in each iteration of ISSA. Inspired by the concept of uniform design, WD is initially adopted to gauge population uniformity, and the threshold acceptance (TA) algorithm is employed to produce an initial population with improved uniformity, hence augmenting the population's diversity and quickening the rate of convergence. Secondly, a macroscopical individual iterative strategy of the producer is adjusted to avoid the population converging to the origin. Then, a dynamic population uniformity affiliation function based on WD is introduced to adjust the population uniformity affiliation function according to the relative amount of change in the global optimum, and the number of danger perceivers is adjusted according to the affiliation function. What is more, a new boundary update strategy is also proposed in ISSA based on population uniformity. By comparing ISSA on 23 standard test functions and recently updated validation function set CEC2022 with the original SSA, and some classical as well as newly developed algorithms, the superiority of the present ISSA is thoroughly confirmed. As an application case, the ISSA algorithm is utilized to solve the path planning problem of the complex environment of unmanned aerial vehicle (UAV) based on threat models by applying it to 2D maps containing circular and polygonal obstacles, as well as 3D maps containing mountain peaks and cylindrical obstacles. The simulation results show that ISSA can find more effective routes through various environments with obstacles.

1. Introduction

With the rapid development of technology, unmanned aerial vehicles (UAVs), due to their high efficiency, low price, ease of management and operation, have become a reliable, convenient and economical option in many fields, such as military (Roberge, Tarbouchi, and Labonte, 2018), disaster management (Alsamhi et al., 2022), logistics and distribution (Ghelichi, Gentili, and Mirchandani, 2022), remote sensing (Liang et al., 2023), and agriculture (Behjati, Noh, Alobaidy, Zulkifley, and Abdullah, 2021), to name a few. The path planning of UAVs is a vital guarantee for them to accomplish their tasks, so many researchers have invested energy in studying how to plan a safe and convenient path (Liu, Shu, Liang, Peng, and Cheng, 2021).

Depending on the level of information available about the environment, UAV path planning problems can be classified into two categories, namely static global planning and dynamic local planning. The global path planning problem holds all the information about the environment, so path planning can be performed directly based on the information held. Local path planning is the real-time collection of environmental information by sensors, which requires high equipment for UAVs (Hu, Zhong, and Wei, 2023). Compared to local path planning, the global UAV path planning problem is a more significant optimization problem subject to several constraints and performance criteria. Until now, there are many algorithms to solve the global path planning, such as Dijkstra method (Atyabi and Powers, 2013), A* algorithm (Jiang, Li, Pan, and Wang, 2020; Zhang, An, Cao, Ouyang,

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and Wang, 2023), artificial potential field algorithm (Sheng, He, Guo, and Li, 2010) and other traditional methods. However, the global path planning is an NP-hard problem containing discontinuities, nonlinearities, and multimodalities. In some complex environments, too many constraints reduce the number of feasible solutions, making UAV path planning more challenging.

In our daily life, there are many discrete optimization problems that are NP-hard problems. With the development of science and technology, the problem to be optimized is becoming more and more complex, and the search space increases exponentially with the size of the problem, and the exact solution method is no longer suitable for solving the NP-hard problem. In order to solve these problems, several algorithm frameworks have been proposed, such as heuristic algorithms, meta-heuristic algorithms, and reinforcement learning algorithms. For example, in the field of logistics and transportation, the introduction of fuzzy layer in the heuristic algorithm provides a feasible method to solve the large-scale NP-hard problem under general uncertainty scenarios (Angel et al., 2023). Grouping problems are a special type of combinatorial optimization problem, and many meta-heuristic methods are used to solve grouping problems, and genetic algorithm (GA) show the best performance in many cases (Octavio, Marcela, Efrén, and Oliver, 2020). The reinforcement learning algorithm can solve the NP-hard problem about graphs, and combine graph isomorphism network and Monte Carlo tree search to solve the combinatorial optimization problem of graphs (Kenshin, Zijian, Issei, and Masashi, 2019). Quantum Approximation Optimization Algorithm (QAOA) has certain advantages in combinatorial optimization problems in NP-hard problems, because there is a beneficial synergy between reinforcement learning and QAOA, and higher quality solutions can be obtained by combining reinforcement learning and recursive QAOA to solve combinatorial optimization problems (Yash, Sofiene, Thomas, and Vedran, 2024). Reinforcement learning has strong generalization ability, but it relies on the interaction process with the environment and takes a long time to calculate. However, heuristics and meta-heuristics do not depend on the interaction of the environment, and even if the optimal solution cannot be obtained in a short time, a suboptimal solution can be obtained. Meta-heuristics are improvements to heuristics that do not depend on the specific structure of the problem, making them more extensive and flexible in solving problems. Some researchers believe that it is not wrong to give up solving the optimal solution in order to shorten the calculation time of the algorithm, that is, to find a suboptimal solution to meet the computational time requirements of the actual task (Mishra et al., 2020). Therefore, meta-heuristics are very advantageous in solving some combinatorial optimization problems in NP-hard problems, such as production scheduling (Gao et al., 2016), task allocation (Yakubu and Murali, 2023), and path planning (Yu, Jiang, Wang, and Li, 2023).

Therefore, the meta-heuristic algorithm has been a powerful way to solve UAV path planning problems. For instance, a technique for planning UAV paths based on the oppositely-learning artificial bee colony algorithm was presented, realizing the collection of more architectural information with fewer images (Chen, Liang, and Meng, 2023). The cooperative path planning problem of UAVs for complicated tasks was solved by using an inventive combination of the task assignment mechanism and the hunter-prey optimization algorithm (Zu, Wang, Liu, and Ge, 2023). The problem of UAV path planning in complex terrain was exploited by introducing an improved nonlinear dynamic inertia weight into the particle swarm optimization (PSO) algorithm (Chu, Yi, and Yang, 2022).

The SSA, proposed based on the foraging and anti-predation behaviors of sparrow populations, is one of the widely recognized meta-heuristic search algorithms in recent years. It is regarded to have good properties of fast convergence speed, good stability, high flexibility, and high solution accuracy, and is highly adaptable to many optimization problems (Fathy, Alanazi, Rezk, and Yousri, 2022; Huang, Zhu, Liu, and Wang, 2022; Qiu and Li, 2022). These advantages make SSA make

positive progress in solving NP-hard problems such as combinatorial optimization, task scheduling, and set partitioning. Usually when solving these NP-hard problems, researchers choose to improve SSA. For example, in order to optimize the power consumption and processing cost at the same time, the improved SSA algorithm is used to solve the scheduling problem (Fei et al., 2022). In order to reduce data redundancy and processing time, and to reduce running memory, improved SSA is often used to solve the optimal or near-optimal feature set in a given dataset (Gad, Sallam, Chakraborty, Ryan, and Abohany, 2022; Jia, Wang, Ahmed, and Ahmed, 2023). SSA needs to be improved in some multi-peak and high-dimensional problems. This is because the initial population is not uniform enough, and the algorithm is easy to fall into the local optimal solution and converge prematurely, resulting in insufficient accuracy and poor solution stability (Yue et al., 2023). To make the algorithm have higher solution accuracy and wider application, many scholars integrate other algorithms or integrate the algorithms with the methods of other disciplines (Li, Guo, Gao, Yang, and He, 2023). For example, the Hammersley low difference sequence strategy was used to produce the initial population, and then the centroid oppositional learning was used to obtain the higher quality of the initial population (Yang, Gao, Zhao, Li, and Gao, 2024). The elite opposition-based learning was used in the process of population initialization to obtain higher quality primarily test populations, preventing the algorithm from converging prematurely. Mutation operations were performed on individuals, and K-means clustering was used with the different probability of mutation operations for different numbers of subpopulations (Leng, Zhang, and Li, 2024). Different individuals adopted different improvement strategies, and dynamic weights and Lévy flight strategies were added to the position update formula of producers to mutate the followers (Yang et al., 2023). The symbiotic stage of SSA's scout with symbiotic organisms search algorithm (SOS) were combined to balance the global search ability and local search ability, increasing the possibility of individuals of approaching the optimal solution (Sun, Si, Ding, Wang, and Xu, 2023).

In this paper, the SSA is improved in several ways based on the theory of uniform design, and the proposed ISSA is successfully applied to UAV path planning problem. The innovations and main contributions of this paper are as follows:

- (1) The idea of uniform design is utilized to generate initial populations with better uniformity. The lattice point method in uniform design is introduced to generate candidate populations, by which the initial populations with good uniformity are generated combined with the uniformity measure WD and TA algorithm;
- (2) The individual iterative strategy of the producer is adjusted at a macro level. This strategy avoids convergence of the population to the origin, broadens the search area, and allows the producer to explore in the early iteration while tending to exploit in the late iteration, which better balances between algorithmic exploration space and exploiting the best capabilities;
- (3) A dynamic population uniformity affiliation function based on WD is introduced, which is adjusted according to the relative change of the global optimum. Then the number of danger perceivers is adjusted according to the affiliation function;
- (4) A new boundary update strategy based on population uniformity is proposed, which adjusts the position of individuals exceeding the boundary according to the current uniformity of the population;
- (5) The improved SSA algorithm is tested using 23 selected classical test functions and recently updated validation functions CEC2022, and its performance was evaluated by comparing it with other meta-heuristic algorithms. The ISSA is also applied to 2D maps containing circular and polygonal obstacles as well as 3D maps containing mountain peaks and cylindrical obstacles to plan safe and convenient UAV paths.

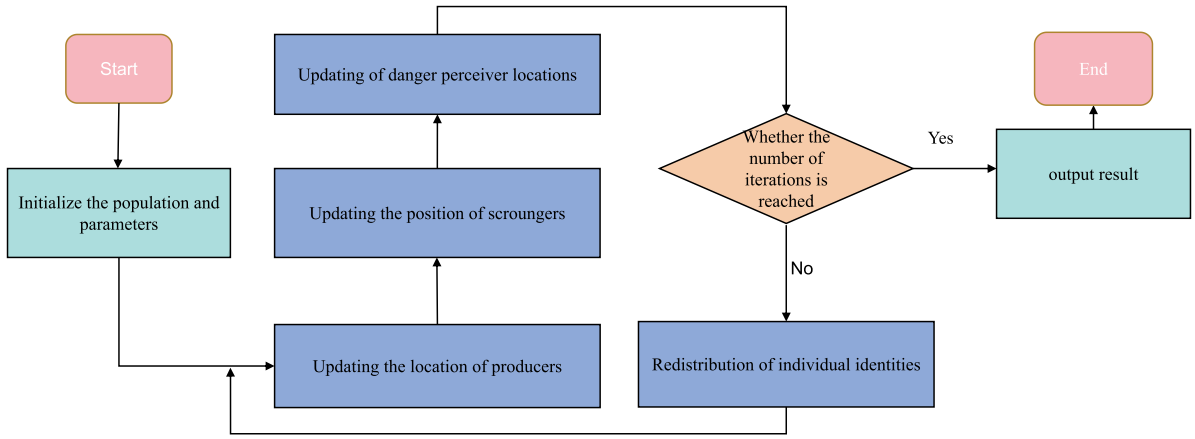


Fig. 1. The specific process of SSA.

The main arrangements of this paper are as follows: Section 2 explains the principles and mathematical formulas of SSA. Section 3 describes the ISSA's optimization strategy. Section 4 shows the results of the ISSA solving of 23 test standard functions and CEC2022. Section 5 analyzes the effectiveness of the ISSA in dealing with complex UAV plan planning problems. Section 6 summarizes the ISSA.

2. Sparrow search algorithm

The original SSA divides sparrows into producers and scroungers based on the amount of food they have in their location and randomly selects some sparrows as danger perceivers. The amount of food at a sparrow's location is also the fitness value of this individual. SSA selects individuals with higher fitness values as producers and the rest as scroungers. In addition, SSA randomly selects some individuals as danger perceivers. Fig. 1 illustrates the main process of SSA.

Producers are responsible for searching to find locations with high fitness values throughout the search area and provide direction for scroungers to search. The number of producers is fixed, typically 10%–30% of the population size. However, the identity of individuals is not fixed, and only individuals with high fitness values can become producers. Influenced by predators, there are two states of discoverers, when there are no predators around, the discoverers will search around themselves, otherwise, all the discoverers will move towards the safe zone. The position of the producer is updated according to Eq. (1).

$$X_i^{t+1} = \begin{cases} X_i^t \cdot \exp(-\frac{i}{\beta_1 \cdot T_{\max}}), & R_2 < ST \\ X_i^t + \beta_2 \cdot L, & R_2 \geq ST \end{cases} \quad (1)$$

where X_i^t is the position of the i^{th} individual at the t^{th} iteration, $T_{\max}$ is the maximum number of iterations, β_1 is a random number uniformly distributed in $[0, 1]$, and β_2 is a random number obeying a standard normal distribution. L is a $1 \times d$ matrix where each element is 1. R_2 represents the alarm value, which follows $[0, 1]$ uniform distribution. The constant ST is the safety value with a value range of $[0.5, 1]$.

The scrounger observes the producer at all times and will adjust its position according to the position of the discoverer, the position of the joiner is updated as shown in Eq. (2):

$$X_i^{t+1} = \begin{cases} \beta_2 \cdot \exp(\frac{X_{worst}^t - X_i^t}{\beta_2^2}), & i > \frac{1}{2}N \\ X_p^{t+1} + |X_i^t - X_p^{t+1}| \cdot L \cdot A^+, & i \leq \frac{1}{2}N \end{cases} \quad (2)$$

where X_{worst}^t is the point with the worst fitness value at the t^{th} iteration, X_p^{t+1} is the optimal producer at the $t+1^{th}$ iteration, and N is the number of populations. A is a $1 \times d$ matrix, which contains only 1 and -1, and these two numbers appear randomly. $A^+ = A^T(AA^T)^{-1}$.

The population randomly selects some of its individuals to act as danger perceivers, which perceive danger and thus fly to other locations. The number of danger perceivers is 10%–20% of the population size. The formula for updating the position of a danger perceiver is shown below:

$$X_i^{t+1} = \begin{cases} X_{best}^t + \beta_3 \cdot |X_i^t - X_{best}^t|, & f_i > f_g \\ X_i^t + \beta_4 \cdot (\frac{X_i^t - X_{worst}^t}{(f_i - f_w) + \rho}), & f_i \leq f_g \end{cases} \quad (3)$$

where X_{best}^t is the globally optimal point, β_3 is the random step control parameter obeying a standard normal distribution; β_4 is a random number obeying the uniform distribution in $[-1, 1]$. f_i is the fitness value of the i^{th} individual, f_g and f_w are the global optimal fitness value and the current worst fitness value, respectively. ρ is a constant, and its role is to prevent the denominator from becoming 0, so it is usually small.

3. The proposed novel optimization method

3.1. Motivation

The superiority of SSA is obvious. Compared with some other classical optimization algorithms, SSA has faster convergence and good optimization-seeking ability. As seen from many experiments, SSA generally performs well for single-peak functions. The performance of SSA is not stable when dealing with multi-peak functions, and SSA is easy to fall into local optimum and cannot get higher convergence accuracy. In addition, the producer tends to converge to the origin when no predator exists. Therefore the algorithm will be relatively poor when dealing with problems where the optimal point is not the origin.

3.2. Population initialization

Like many classical algorithms, SSA generates the initial population in a pseudo-randomized manner, potentially leading to an uneven distribution of the initial population. The prior results have indicated that the discovery of the global optimal solution depends largely on the initial starting point of the meta-heuristic algorithm, and improving the uniformity of the initial population can enhance the diversity of the population and accelerate the rate of convergence (Yang, Deb, Zhao, Fong, and He, 2018). Most of the improvements of the initial population are using a chaotic number generator to generate the initial population, however, chaotic sequences are too sensitive to the initial values and parameters, which need to be chosen differently for different problems. In addition, chaotic sequences may be attracted to several fixed points, which can seriously affect the algorithm results (Yang et al., 2023). Therefore, in this paper, we choose to use the idea of

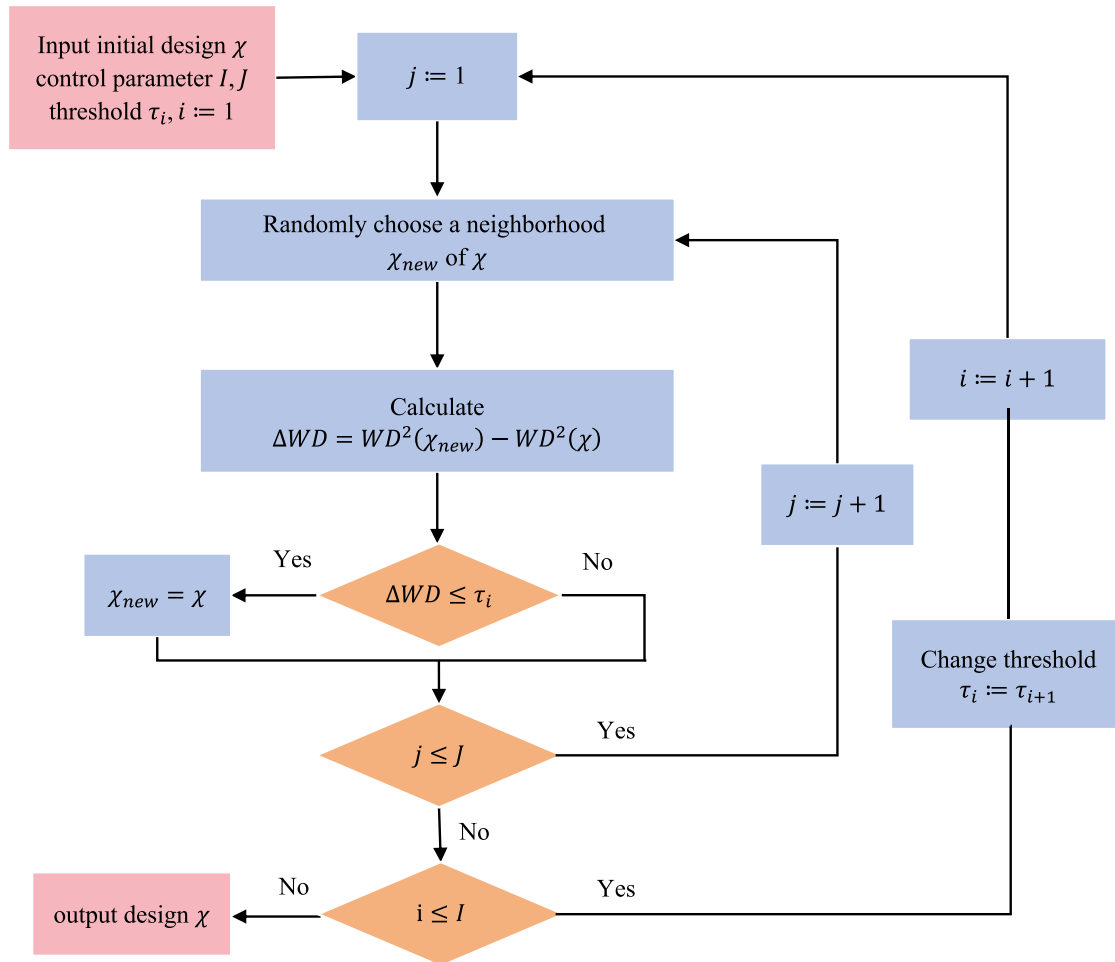


Fig. 2. Flowchart of TA algorithm.

uniform design (Fang, Lin, Winker, and Zhang, 2000) to generate a more uniform initial population.

In addition, many scholars have proposed new strategies for generating initial populations when improving optimization algorithms, but rarely have we seen formulas for measuring population uniformity. Therefore, we introduce discrepancy as a measure of population uniformity. In optimization problems, there are 30-dimensional, 50-dimensional as well as 100-dimensional problems. Not all the discrepancies satisfy the smaller dimensional catastrophe problem, such as the common L_2 -discrepancy, so we choose WD, which satisfies the smaller dimensional catastrophe problem, as the uniformity measure (Matsumoto, Saito, and Matoba, 2013). The WD formula is as follows:

$$WD(X) = \left[\frac{1}{n} \left(\frac{1}{2} \right)^s - \left(\frac{4}{3} \right)^s + \frac{2}{n^2} \sum_{i=1}^{n-1} \sum_{k=i+1}^n \prod_{j=1}^s \frac{3}{2} - |x_{ij} - x_{kj}| + |x_{ij} - x_{kj}|^2 \right]^{\frac{1}{2}} \quad (4)$$

where $X = (x_{ij})_{n \times s}$.

We use the good lattice point designs to generate a preparatory population, utilizing WD as the uniformity measure, and the TA algorithm to select the designs with good uniformity as the initial population.

The steps regarding the good lattice point designs are as follows:

- (1) Given n , define the set of positive integers $H_n = \{h : h \text{ and } n \text{ are coprime, } h < n\} \equiv \{h_1, h_2, \dots, h_m\}$, where the greatest common divisor of h_j and n is 1, and $h_1 < h_2 < \dots < h_m < n$;

Table 1

The mean WD obtained by 30 replicates of the experiment for the three methods.

Dim	Uniform design	Logistic mappings	Chaotic mappings
10	0.7595	1.4743	1.3222
20	7.187	10.1463	10.9804
30	59.3551	68.7302	71.3733

- (2) For each $h \in H_n$, generate a U -shaped design $\chi^h = (x_{ij})$ as follows:

$$x_{ij} = ih_j \pmod{n} \quad (5)$$

where $\pmod{n}$ is the congruence operation;

- (3) Select the dim column in χ with the best uniformity for the dimension of the optimization.

To get the design with the best uniformity, the size of the deviation of $\binom{m-1}{dim-1}$ designs need to be calculated, for example, when $n = 53$, $dim = 30$, $\binom{52}{29} \approx 9.1224e^{36}$. To further reduce the computational complexity, TA is used to construct a uniformly good initial population. The flowchart of TA can be shown in Fig. 2, which is explained as follows:

- (1) χ_{new} is a neighborhood of χ obtained by swapping some two elements of a column in χ ;

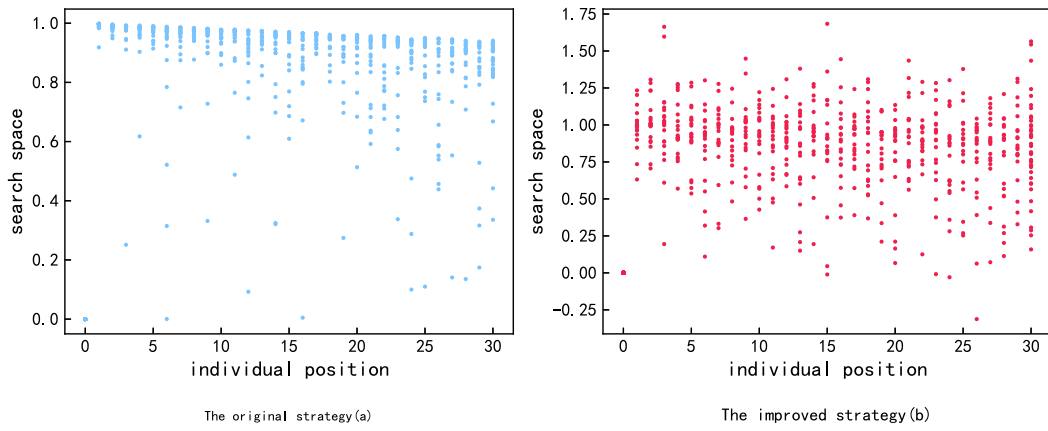


Fig. 3. The scope of the producer's search ($R_2 < ST$).

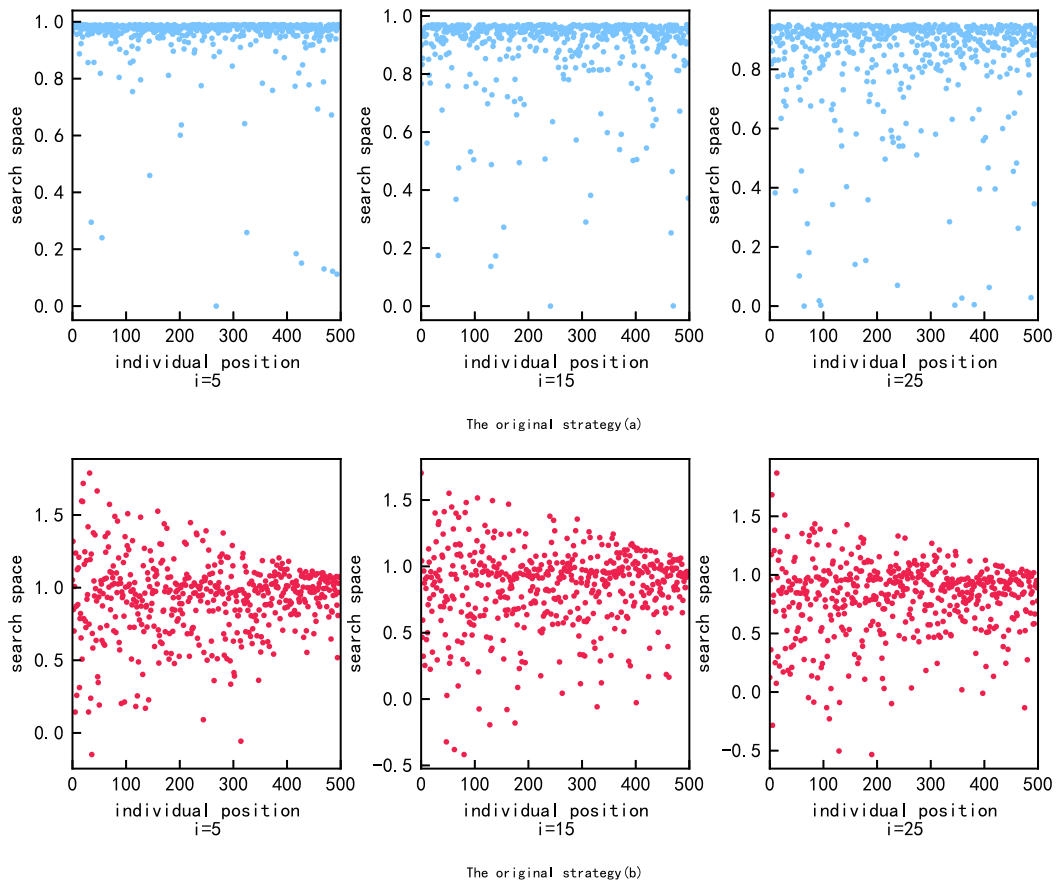


Fig. 4. Trends in the position of producers ($R_2 < ST$).

- (2) It is important to choose the threshold τ_i appropriately, and some scholars have suggested $\tau_1 > \tau_2 > \dots > \tau_I = 0$;
- (3) Set I for the number of times the threshold is lowered and set J to control the number of searches per threshold.

It should also be explained that for experiments, it is common to choose populations of 30, 50, or 100, so when generating the preparatory populations using the good lattice point designs, set the number to a prime number larger than the population, and then remove the extra columns at the end.

It is common to use chaotic mapping to improve initial population uniformity in many other improved algorithms, such as cubic sequence mapping as well as logistic sequence mapping. WD is used as a measure of uniformity to compare the ability of the three methods to improve population uniformity. To fairly compare the ability of the three methods, the number of populations is set to be 50, and when the dimensions are 10, 20, and 30, each method independently and repeatedly generates 30 initial populations, and the average WD values of the three methods are obtained. The results are shown in Table 1. Obviously, the WD value of the initial population constructed in this

paper is lower than that of the cubic sequence mapping and the logistic sequence mapping.

3.3. Modifying the producer location update strategy from a macro perspective

For the producer, when no danger is encountered, the individual searches for food around itself, and when danger is encountered, the individual jumps out of the current position. However, from Fig. 3(a), it can be seen that when no danger is encountered, the individual's position has a tendency to converge to the origin, and the individual is merely approaching the origin and giving up looking in the opposite direction to the origin. In addition, as can be seen from Fig. 4(a), individuals do not show different iterative strategies at different iterative periods, only the positions of different individuals will differ. Ideal producer explores around itself rather than being confined to the space between its own position and the origin. They tend to explore in the early part of the iteration while they tend to exploit in the later part of the iteration. To address these two issues, we improve the formula as follows:

$$X_i^{t+1} = X_i^t \cdot \left(\exp\left(-\frac{i}{\alpha \cdot T_{\max}}\right) + r_1 \cdot (2 \cdot r_2 - 1) - (2^{r_1 \cdot (2 \cdot r_2 - 1)} - 1) \cdot \frac{t}{T_{\max}} \right), R_2 < ST \quad (6)$$

where r_1 and r_2 are uniform distributions with an interval of $[0, 1]$. As shown in Fig. 3, compared to the original SSA producer, the modified discoverer no longer searches between itself and the origin, but searches in its own neighborhood. The macroscopic effect of the modification is shown in Fig. 4(b), where the search range of the individual shrinks as the number of iterations increases, which allows the algorithm to perform extensive search in the pre-iteration period, thus enhancing the algorithm's ability to jump out of the local optimum. As the algorithm iterates, the producer tends to exploit more. Therefore, the modified producer better balances the exploration and exploitation capabilities of the algorithm, which can effectively prevent the algorithm from falling into local optimality in the early stage and accelerate convergence in the later stage.

When danger is encountered, the direction and displacement of the individual's movement are shown in Fig. 5, which obviously does not take into account that different problems face different search spaces. When the search space is small, the position of the updated individual is likely to be directly outside the search space. In order not to waste the ability of the producer, the search formula for $R_2 \geq ST$ is improved as follows:

$$X_i^{t+1} = X_i^t + \beta_2 (X_{rand}^t - X_i^t), R_2 \geq ST \quad (7)$$

where X_{rand}^t is the position of a random individual with the top 50% fitness value at the t^{th} iteration, and β_2 is in $(0, 1]$. Individuals randomly move in the direction of individuals with higher fitness values in the population. This allows individuals to explore the space without going beyond the search space. At the same time, it also increases the randomness of the population and reduces the possibility of falling into the local optimum.

3.4. Mechanisms for dynamic regulation of the number of danger perceivers

When $f_i > f_g$, the sparrow is very vulnerable to predators, and danger perceivers will approach the safe zone in the original SSA. However, when the algorithm falls into local optimization, too many dangers approaching the safe zone will only make the algorithm fall into the "stagnation period", so this paper introduces the WD in uniform design to measure the uniformity of the population. From Eq. (4), it is easy to know that $WD(X_{n \times s})_{Max} = (-\frac{4}{3})^s + (\frac{3}{2})^s)^{1/2}$, the smaller the value of WD, the better the uniformity, so we use ϕ to quantify the uniformity of the population:

$$\phi = 1 - \frac{WD(X_{n \times s})}{WD(X_{n \times s})_{Max}} \quad (8)$$

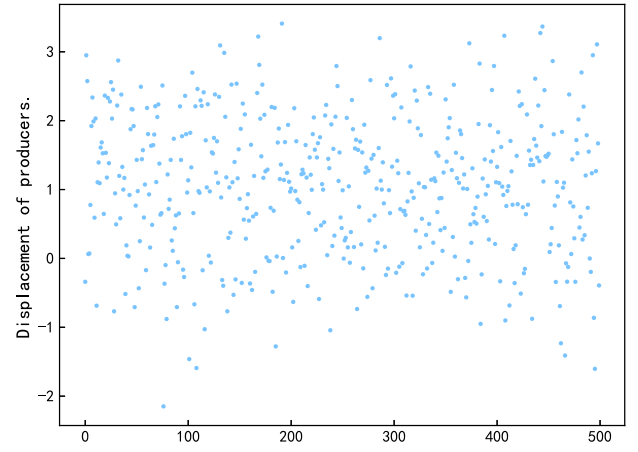


Fig. 5. Producer displacement ($R_2 \geq ST$).

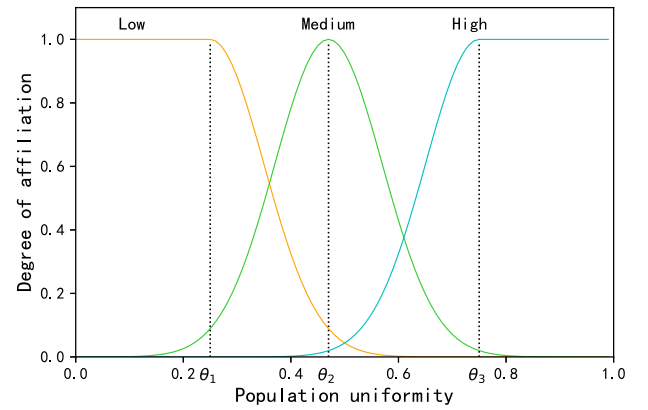


Fig. 6. Affiliation of population uniformity ϕ .

(see Fig. 6).

When the uniformity of the population is high, the individuals are scattered in the search space, and then increasing the number of danger perceivers is favorable to the convergence of the population. When the uniformity of the population is low, the population gathers together, then reducing the number of danger perceivers is not only helpful to prevent the algorithm from falling into the local optimum, but also can reduce the consumption of computing resources.

The uniformity of the population can be categorized into three states through the affiliation function $A(\phi) : A(\phi) = \{1, 2, 3\}$. They represent three levels of uniformity of the population: "Low", "Medium" and "High". When the uniform level of the population is different, the proportion of the number of danger perceivers SD also varies, as follows:

$$SD = \begin{cases} 0.1, & A(\phi) = 1 \\ 0.15, & A(\phi) = 2 \\ 0.2 & A(\phi) = 3 \end{cases} \quad (9)$$

In the process of seeking the optimal solution, if the population uniformity affiliation mechanism has been fixed, the number of danger perceivers is adjusted according to the unchanging mechanism. This is not an ideal mechanism to regulate the number. This is because it is possible to encounter a situation where the value of WD is low, but does not fall into the local optimum. At this time, the number of danger perceivers should be at a relatively high state, rather than lowering the number of danger perceivers because of low uniformity. In order to adjust the number of danger perceivers more objectively, we introduce the global optimal relative change γ and judge whether the population

falls into the local optimal according to γ and the state of population uniformity. The expression for γ is shown:

$$\gamma = \frac{f_{best}^t - f_{best}^{t-1}}{f_{best}^{t-1}} \quad (10)$$

At the t^{th} iteration, if the population uniformity state is “Low” and $\gamma < 0.01$, that is, the fitness value does not change by more than 1% after this iteration and the current individuals are all clustered together. We believe that there is a possibility that the population will stagnate at the local optimum. At this time, the number of danger perceivers is not conducive to population optimization but wastes computing resources. This is not beneficial for population optimization and wastes computational resources. Therefore, we adjust the affiliation mechanism by shifting $\theta^t = (\theta_1, \theta_2, \theta_3)^T$ to the right to get θ^{t+1} at the $(t+1)^{th}$ iteration, thus obtaining a new mechanism for the number of danger perceivers for population uniformity. θ^{t+1} is shown as follows:

$$\theta^{t+1} = \theta^t + \eta \quad (11)$$

where η is a constant greater than 0.

If $\gamma \geq 0.1$, it shows that the global optimum changes during the iteration, so it can be assumed that not only is the population not stagnant at the local optimum, but that it is generally possible to find a better position near the current global optimum. Therefore, more dangers are needed at this point to move closer towards the global optimum. Therefore, θ^t will be shifted to the left to obtain θ^{t+1} , and the expression for h_1 is shown as follows:

$$\theta^{t+1} = \theta^t - \eta \quad (12)$$

If $0.01 \leq \gamma < 0.1$, the population is considered to be in normality, then θ^t does not move and $\theta^{t+1} = \theta^t$.

3.5. A new boundary control strategy

In the original SSA, individuals exceeding the boundary are handled by the traditional boundary processing mechanism, that is, individuals exceeding the boundary are given a fixed value. However, assigning a fixed value to an individual not only fails to utilize the individual's optimization ability, but also leads to a decrease in population diversity. Some scholars will improve the traditional boundary control strategy. For example, Wu et al. randomly move individuals beyond the boundary to the neighborhood of the global optimum solution (Wu et al., 2023), which not only improves the population diversity but also facilitates individuals to exploit better locations compared to the traditional approach. In Section 3.4, we establish the population uniformity affiliation mechanism, which determines different boundary processing mechanisms based on the degree of uniformity of the population. When the population uniformity is low, individuals are clustered near the optimal value, and setting the new positions of the individuals that exceed the boundary near the global worst value and away from the global optimal value can improve the diversity of the population, which is helpful to enhance the ability of the algorithm to jump out of the local optimum. When the population uniformity is medium, the new position is then randomly generated between the global optimum and the global worst value, ensuring that the new position does not exceed the boundary. When the population uniformity is high and the individuals are more dispersed, the new position is set near the optimal value, which is conducive to enhancing the exploitation ability of the population. Therefore the new position is formulated as follows:

$$X_i^{t+1} = \begin{cases} \varpi_1 X_{worst}^t + (1 - \varpi_1) X_{best}^t, & A(\phi) = 1 \\ \varpi_2 X_{worst}^t + (1 - \varpi_2) X_{best}^t, & A(\phi) = 2 \\ X_{best}^t + X_{distance}, & A(\phi) = 3 \end{cases} \quad (13)$$

where ϖ_1 is a random number between $[0.8, 1)$, and ϖ_2 and ϖ_3 are random numbers between $(0, 1)$, $X_{b_2}^t$ and $X_{b_3}^t$ are the suboptimal and third best solutions in the population, and $X_{distance} = \varpi_3 |X_{best}^t - X_{b_2}^t| + (1 - \varpi_3) |X_{best}^t - X_{b_3}^t|$.

Table 2

Setting of parameter values for the selected optimization algorithm.

Algorithm	Parameter settings	Ref.
PSO	$c_1 = c_2 = 1.5, \omega = 0.9$	(Kennedy and Eberhart, 1995)
DE	$f = 0.2, cr = 0.2$	(Storn and Price, 1997)
GWO	$a = [2, 0]$	(Seyedali, Seyed Mohammad, and Andrew, 2014)
WOA	$a = [2, 0], a2 = [-2, -1], b = 1$	(Mirjalili and Lewis, 2016)
BES	$a = 10, R = 1.5, \alpha = 2, c1 = 2, c2 = 2$	(Alsattar, Zaidan, and Zaidan, 2020)
HHO	none	(Heidari et al., 2019)
SMA	$z = 0.3$	(Li, Chen, Wang, Heidari, and Mirjalili, 2020)
SSA	$PD = 20\%, SD = 15\%, ST = 0.8$	(Xue and Shen, 2020)
AO	$\alpha = 0.1, \delta = 0.1, r_1 = 10$	(Abualigah et al., 2021)
ISSA	$PD = 20\%, SD = [10\%, 20\%], ST = 0.8$	

Table 3

Unimodal benchmark functions.

Function	Dim	Range	f_{min}
$F_1(x) = \sum_{i=1}^n x_i^2$	30	$[-100, 100]$	0
$F_2(x) = \sum_{i=1}^n x_i + \prod_{i=1}^n x_i $	30	$[-10, 10]$	0
$F_3(x) = \left(\sum_{i=1}^n x_i \right)^2$	30	$[-100, 100]$	0
$F_4(x) = \max \{ x_i , 1 \leq i \leq n \}$	30	$[-100, 100]$	0
$F_5(x) = \sum_{i=1}^{n-1} \left[100(x_{i+1} - x_i^2)^2 + (x_i - 1)^2 \right]$	30	$[-30, 30]$	0
$F_6(x) = \sum_{i=1}^n \left(x_i + 0.5 \right)^2$	30	$[-100, 100]$	0
$F_7(x) = \sum_{i=1}^n i x_i^4 + \text{random}[0, 1)$	30	$[-1.28, 1.28]$	0

4. Experiments and analysis

4.1. Results comparisons using 23 test functions

To verify the convergence speed, convergence accuracy, and robustness of ISSA, we use the ISSA algorithm to perform numerical simulation experiments on 23 standard test functions (including unimodal benchmark function, multimodal benchmark function, and fixed-dimension multimodal benchmark functions), and compare the optimization results with those of some classical algorithms in recent years (see Tables 3–5). In this paper, the following algorithms were selected for comparative experiments: PSO, Differential Evolutionary (DE), Gray Wolf Optimizer (GWO), Whale Optimization Algorithm (WOA), Bald Eagle Search (BES), Harris Hawks Optimization (HHO), Slime Mould Algorithm (SMA), SSA, and Aquila Optimizer (AO). Table 2 shows the parameters of these algorithms, which are the recommended values of the original algorithms.

To fairly compare the optimization ability of these algorithms, the maximum number of iterations is set to 500, and the number of individuals participating in the experiment for each algorithm is 30, considering the randomness of the meta-heuristic algorithms, each algorithm is subjected to 30 independent repetitions of the experiment. To make the results more readable and objective, we used two statistical indicators: mean (Mean) and standard deviation (Std). Under the same test function, the mean indicates convergence accuracy, and the standard deviation indicates the robustness of the algorithm. Based on these 23 test functions, the comparison results of these algorithms are shown in Table 6, and the convergence curves are shown in Fig. 7. To show the convergence curves of each algorithm more reasonably, the convergence curves are chosen to be the data of the results ranked 15 in 30 experiments of each algorithm. To show the convergence accuracy

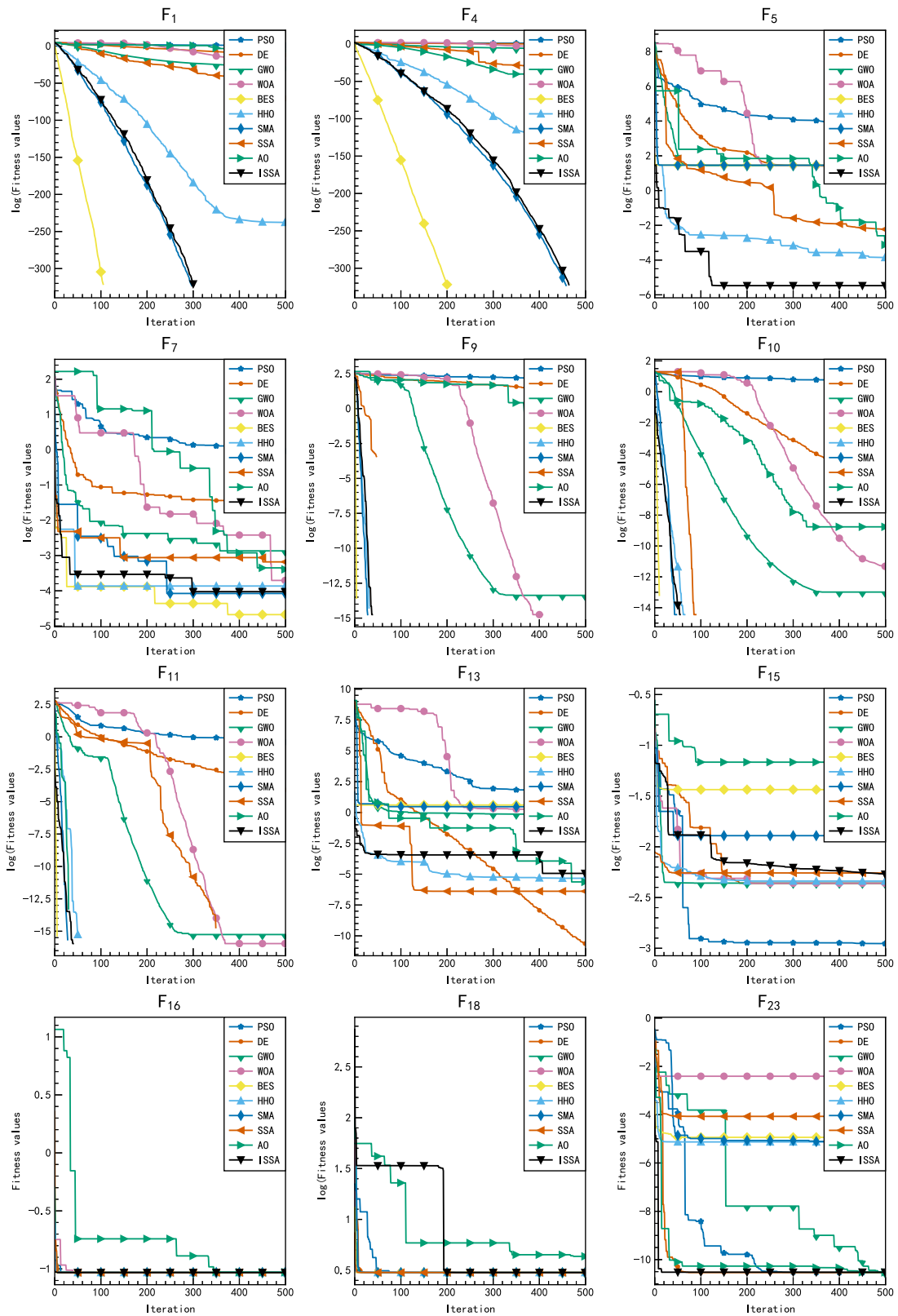


Fig. 7. Convergence curves of ISSA and other optimization algorithms on standard test functions.

and robustness of each algorithm at the same time, we plotted the box-and-line diagrams of the test functions, as shown in Fig. 8. In addition, this paper uses the Wilcoxon rank sum test to demonstrate the ISSA's ability to find optimization.

Table 6 analyzes the optimization ability of the algorithm. For these 23 test functions. It can be seen that the ISSA has the best average ranking of 2.83 across all tested functions. It can find the optimal values of F_1 , F_2 , F_3 , F_4 , F_9 , F_{10} , and F_{11} in each experiment. The

Table 4
Multimodal benchmark functions.

Function	Dim	Range	$f_{\min}$
$F_8(x) = \sum_{i=1}^n -x_i \sin(\sqrt{ x_i })$	30	[-500,500]	-4.18.9892n
$F_9(x) = \sum_{i=1}^n [x_i^2 - 10 \cos(2\pi x_i) + 10]$	30	[-5.12,5.12]	0
$F_{10}(x) = -20 \exp(-0.2 \sqrt{\frac{1}{n} \sum_{i=1}^n x_i^2} - \exp(\frac{1}{n} \sum_{i=1}^n \cos(2\pi x_i))) + 20 + e$	30	[-32,32]	0
$F_{11}(x) = \frac{1}{4000} \sum_{i=1}^n x_i^2 - \prod_{i=1}^n \cos(\frac{x_i}{\sqrt{i}}) + 1$	30	[-600,600]	0
$F_{12}(x) = \frac{\pi}{n} \{10 \sin(\pi y_1) + \sum_{i=1}^{n-1} (y_i - 1)^2 [1 + 10 \sin^2(\pi y_{i+1})] + (y_n - 1)^2\} + \sum_{i=1}^n u(x_i, 10, 100, 4)$ $y_i = 1 + \frac{x_i + 1}{4}$ $u(x_i, a, k, m) = \begin{cases} k(x_i - a)^m, & x_i > a \\ 0, & -a < x_i < a \\ k(-x_i - a)^m, & x_i < -a \end{cases}$	30	[-50,50]	0
$F_{13}(x) = 0.1 \{ \sin^3(3\pi x_1) + \sum_{i=1}^n (x_i - 1)^2 [1 + \sin^2(3\pi x_i + 1)] + (x_n - 1)^2 [1 + \sin^2(2\pi x_n)] \} + \sum_{i=1}^n u(x_i, 5, 100, 4)$	30	[-50,50]	0

Table 5
Fixed-dimension multimodal benchmark functions.

Function	Dim	Range	$f_{\min}$
$F_{14}(x) = \left(\frac{1}{500} + \sum_{j=1}^{25} \frac{1}{j + \sum_{i=1}^{25} (x_i - a_{ij})^6} \right)$	2	[-65,65]	1
$F_{15}(x) = \sum_{i=1}^{11} \left(a_i - \frac{x_i(b_i^2 + b_i x_2)}{b_i^2 + b_i x_3 + x_4} \right)^2$	4	[-5,5]	0.0003
$F_{16}(x) = 4x_1^4 - 2.1x_1^4 - \frac{1}{3}x_1^6 + x_1x_2 - 4x_2^2 - 4x_2^4$	2	[-5,5]	-1.0316
$F_{17}(x) = \left(x_2 - \frac{5.1}{4\pi^2} x_1^2 + \frac{5}{\pi} x_1 - 6 \right)^2 + 10 \left(1 - \frac{1}{8\pi} \right) \cos x_1 + 10$	2	[-5,5]	0.398
$F_{18}(x) = [1 + (x_1 + x_2 + x_3)^2 (19 - 14x_1 + 3x_1^2 - 14x_1 + 6x_1x_1 + 3x_2^2)] \times [30 + (2x_1 - 3x_2)^2 \times (18 - 32x_1 + 12x_1^2 + 48x_1 - 36x_1x_1 + 27x_2^2)]$	2	[-2,2]	3
$F_{19}(x) = -\sum_{i=1}^4 c_i \exp \left(-\sum_{j=1}^3 a_{ij} (x_j - p_{ij})^2 \right)$	3	[0,1]	-3.86
$F_{20}(x) = -\sum_{i=1}^4 c_i \exp \left(-\sum_{j=1}^6 a_{ij} (x_j - p_{ij})^2 \right)$	6	[0,1]	-3.32
$F_{21}(x) = -\sum_{i=1}^5 [(X - a_i)(X - a_i)^T c_i]^{-1}$	4	[0,10]	-10.1532
$F_{22}(x) = -\sum_{i=1}^7 [(X - a_i)(X - a_i)^T c_i]^{-1}$	4	[0,10]	-10.4028
$F_{23}(x) = -\sum_{i=1}^{10} [(X - a_i)(X - a_i)^T c_i]^{-1}$	4	[0,10]	-10.5363

second and third highest ranked algorithms are HHO and SMA, with average rankings of 3.7 and 4.87, respectively. The two worst-ranked algorithms are PSO and WOA, both of which have an average rank greater than 6. In conjunction with Fig. 8, in addition to the 7 test functions where the optimal values can be found, the ISSA has the smallest standard deviations on F_5 , F_{13} , F_{16} , F_{20} , F_{21} , F_{22} , and F_{23} . The ISSA has the smallest standard deviation on a total of 14 test functions, followed by the SMA, which has the smallest standard deviation on a total of 9 test functions. The ISSA is the most stable compared to other algorithms in the experiment.

Let us analyze the convergence speed of the algorithms through Fig. 7. F_1 , F_4 , F_5 and F_7 are unimodal benchmark functions. Regarding F_1 and F_4 , the fastest convergence speed is BES, followed by SMA and ISSA, and all three algorithms find the optimal value of the function. HHO also converges very fast, and although it does not find the optimal value, it converges faster than the other six algorithms. Regarding F_5 , from the iterative curve, it can be intuitively seen that ISSA has quickly converged to the global optimum. HHO is second only to ISSA in terms of convergence speed and convergence accuracy. Combined with Table 6, the ISSA has the highest convergence accuracy at F_1 , F_2 , F_3 , F_4 and F_5 . In addition, the ISSA has the best stability on F_1 , F_2 , F_3 and F_4 . From the perspective of convergence speed, convergence accuracy and stability, ISSA is the most competitive algorithm in the single-peak benchmark function. F_9 , F_{10} , F_{11} and F_{13} are multimodal benchmark functions, and on F_9 , the best performer is BES, which has a very fast convergence speed, followed by HHO, SMA, SSA, and

ISSA. These algorithms find the optimal value of the function after less than 50 iterations, the optimum was found. The five algorithms that performed very well on F_9 performed equally well on F_{10} . The slowest converging algorithm, SSA, found the optimum in less than 100 iterations. On F_{11} , BES, HHO, SMA, and ISSA all converge very fast, followed by WOA and SSA. On F_{12} , the fastest convergence is DE, followed by SSA and HHO, and the improved algorithm of this paper, ISSA, ranks fifth in terms of convergence speed. On F_{13} , the ISSA did not perform well, and the ISSA fell into the local optimal solution prematurely, resulting in low solution accuracy. Combined with Table 6, the performance of the ISSA on the multimodal benchmark function shows that although the ISSA does not have a high accuracy in solving F_{13} , the convergence speed, convergence accuracy and stability of the ISSA in solving F_9 , F_{10} , F_{11} and F_{12} are excellent. F_{15} , F_{16} , F_{18} and F_{23} are fixed-dimension multimodal benchmark functions, and on F_{15} , the best convergence speed and convergence accuracy is achieved by PSO. AO, BES, and SMA converge slowly and the rest of the algorithms have similar convergence speeds. On F_{16} and F_{18} , these 10 algorithms have similar convergence speeds, and only AO is worse. On F_{23} , the fastest convergence speed is ISSA, followed by PSO, DE, and AO, while WOA converges fast with poor convergence accuracy. As can be seen from Fig. 7, the ISSA converges quickly and accurately at F_{16} , F_{18} and F_{23} . Combined with Table 6, the ISSA performs well in solving fixed-dimensional multimodal benchmark functions. Taken together, ISSA, BES, HHO, and SMA converge faster than the other six algorithms on the 23 tested functions.

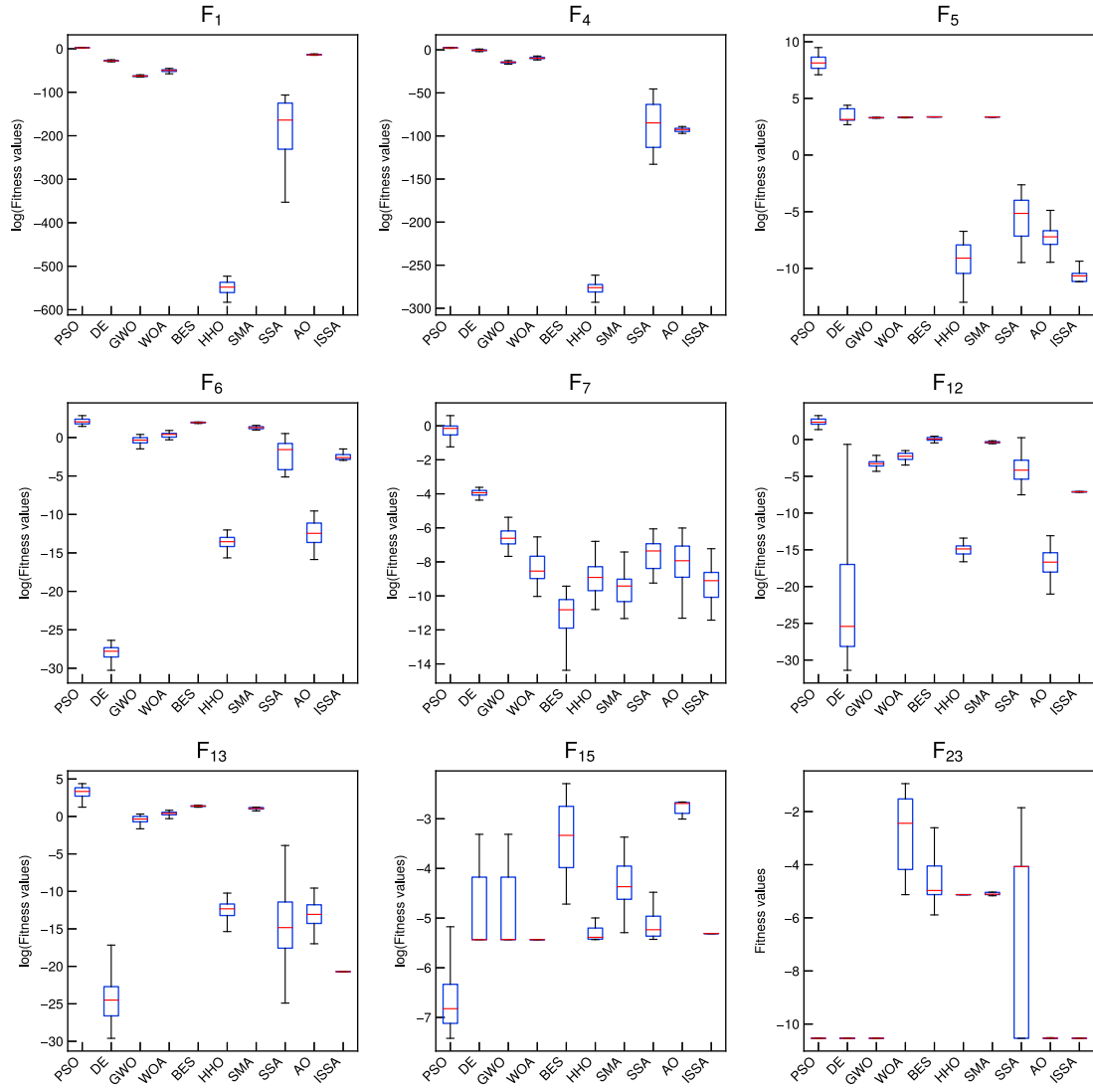


Fig. 8. Boxplot of ISSA and other optimization algorithms on the standard test function.

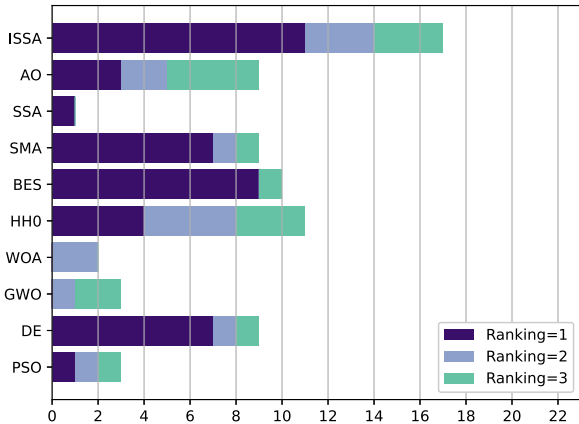


Fig. 9. Histogram of the ordering of algorithms such as ISSA on 23 standardized test functions.

Regarding the robustness of the algorithms, it can be seen from Table 7 that the 10 algorithms with the best average ranking are ISSA, followed by BES, HHO, and the worst average ranking is PSO. Fig. 8 shows the box line plots of the ten algorithms, and it can be easily seen

that ISSA is the most robust and has the highest convergence accuracy on the test functions F_1 , F_2 , F_5 , and F_{23} . On F_6 , F_7 , F_{12} , and F_{15} , ISSA is more robust, but the convergence accuracy is not the best. Not all algorithms with both high convergence accuracy and robustness will be available on all test functions, such as F_{12} and F_{13} , where DE has the highest convergence accuracy but weak robustness.

In conclusion, analyzing the convergence accuracy, convergence speed, and robustness together, the algorithm in this paper has the best performance, which can balance the ability to explore space and mine the optimal value. It is followed by HHO and BES with good performance such as convergence accuracy (see Fig. 9).

As can be seen from Table 6, the ISSA ranks first in five test functions, 4th on F_6 and 3rd on F_7 on the unimodal test functions F_1 - F_7 . Compared to the other nine algorithms, the ISSA performed best on the unimodal test function. The original SSA has an average ranking of 5.28 on the 7 unimodal test functions, it can be seen that the ISSA's improvements in this type of test function are effective. On the multimodal test functions F_8 - F_{13} , the ISSA ranks first on four test functions and third on F_{12} . Compared to the original SSA, the improvement effect is obvious. On the fixed-dimension multimodal test functions F_{14} - F_{23} , the ISSA ranks in the top three on only six functions. Poor performance on F_{14} , F_{17} , F_{19} , F_{20} . SSA has an average ranking of 6.9 on F_{14} - F_{23} , the ISSA has an average ranking of 3.6. Out of the 10 algorithms, the best performer was DE with an average ranking of 2.5,

Table 6

Experimental results of ISSA and other optimization algorithms on standard test functions.

Function	Index	ISSA	PSO	DE	GWO	WOA	BES	HHO	SMA	SSA	AO
F_1	Mean	0	8.9849E+00	2.4972E-12	1.3619E-27	2.8084E-21	0	3.4439E-229	0	1.5173E-34	1.6807E-06
	Std	0	3.2624E+00	5.6724E-12	2.3107E-27	6.0284E-21	0	0.0000E+00	0	8.1707E-34	1.8091E-06
F_2	Mean	0	1.7338E+01	2.8620E-06	9.9161E-17	8.9451E-15	0	1.0770E-116	0	2.3687E-24	2.6286E-03
	Std	0	5.3948E+00	7.8386E-06	6.0837E-17	1.2874E-14	0	5.7510E-116	0	9.9457E-24	2.3548E-03
F_3	Mean	0	5.4200E+02	1.9413E+01	4.7391E-06	7.5138E-03	0	2.1142E-140	0	5.4263E-29	4.9017E-03
	Std	0	2.0059E+02	1.8960E+01	1.0797E-05	2.0928E-02	0	1.1385E-139	0	2.9160E-28	7.1180E-03
F_4	Mean	0	9.4283E+00	7.0691E-01	7.5926E-07	1.7262E-04	0	8.2987E-116	0	1.1159E-21	2.5049E-40
	Std	0	2.4982E+00	5.0361E-01	7.4153E-07	3.2511E-04	0	4.4514E-115	0	5.8186E-21	4.4415E-40
F_5	Mean	3.6290E-05	4.0643E+03	3.5817E+01	2.7044E+01	2.7839E+01	2.8980E+01	2.4485E-04	2.8778E+01	2.5639E-02	1.6849E-03
	Std	3.2718E-05	2.6098E+03	2.5940E+01	6.6121E-01	7.0206E-01	1.4258E-02	2.9073E-04	2.0584E-01	5.6642E-02	2.9149E-03
F_6	Mean	9.7039E-02	8.7966E+00	1.0256E-12	7.1691E-01	1.4296E+00	6.9631E+00	1.6086E-06	3.6309E+00	4.3541E-01	1.3023E-05
	Std	5.5194E-02	3.4858E+00	8.2724E-13	3.5922E-01	5.0943E-01	5.0700E-01	1.3296E-06	6.0420E-01	4.5065E-01	1.8786E-05
F_7	Mean	1.4879E-04	8.1528E-01	1.9958E-02	1.5240E-03	3.4958E-04	2.5943E-05	2.1616E-04	1.0823E-04	9.3554E-04	5.3145E-04
	Std	1.5365E-04	3.3754E-01	4.4213E-03	8.4593E-04	3.6151E-04	2.1997E-05	2.3689E-04	1.2317E-04	8.3434E-04	5.3074E-04
F_8	Mean	-3.1763E+03	-6.3845E+03	-9.4977E+03	-5.9361E+03	-6.9694E+03	-4.5518E+03	-1.1995E+04	-7.8797E+03	-7.2222E+03	-1.2325E+04
	Std	8.5937E+02	7.8603E+02	2.2454E+03	1.0962E+03	2.9709E+02	6.5910E+02	1.2115E+03	6.0407E+02	9.7511E+02	6.8266E+02
F_9	Mean	0	1.3180E+02	3.7872E+01	2.6523E+00	1.1842E-16	0	0	0	0	1.6619E-04
	Std	0	2.3204E+01	4.1190E+01	3.7346E+00	4.4310E-16	0	0	0	0	2.3777E-04
F_{10}	Mean	0	5.6611E+00	3.1044E-02	1.0196E-13	1.0267E-11	0	0	0	3.5527E-16	1.2140E-08
	Std	0	7.9936E-01	1.6717E-01	1.3977E-14	1.6253E-11	0	0	0	1.4062E-15	2.9295E-08
F_{11}	Mean	0	4.2172E-01	9.9108E-05	4.3119E-03	7.4015E-17	0	0	0	2.8401E-03	0
	Std	0	1.9609E-01	6.8346E-05	5.3304E-03	7.2141E-17	0	0	0	1.5295E-02	0
F_{12}	Mean	8.1220E-04	1.3718E+01	3.4560E-02	4.2950E-02	1.6886E-01	1.1090E+00	4.4123E-07	6.8132E-01	5.6612E-02	2.3460E-07
	Std	2.8705E-05	1.0231E+01	1.0132E-01	2.2696E-02	1.8256E-01	2.7288E-01	3.6972E-07	8.6334E-02	6.6889E-02	4.0976E-07
F_{13}	Mean	1.6678E-08	3.2421E+01	2.6252E-03	7.6311E-01	1.4832E+00	3.9589E+00	1.3376E-03	2.8926E+00	2.6497E-03	8.9958E-06
	Std	4.6948E-08	2.1004E+01	1.4137E-02	3.3529E-01	4.3515E-01	2.8940E-01	4.9961E-03	4.3804E-01	1.3845E-02	1.5801E-05
F_{14}	Mean	2.5543E+00	3.7895E+00	9.9800E-01	4.0679E+00	1.3506E+00	5.9577E+00	2.2476E+00	1.7197E+00	4.7152E+00	2.4945E+00
	Std	3.9679E+00	2.6891E+00	0.0000E+00	3.9372E+00	5.4316E-01	4.2196E+00	2.0864E+00	1.9401E+00	4.3912E+00	1.4612E+00
F_{15}	Mean	6.2927E-03	2.0180E-03	1.0832E-02	1.2718E-02	4.3697E-03	4.1760E-02	6.9092E-03	1.4366E-02	9.7605E-03	6.1088E-02
	Std	6.0406E-03	2.8304E-03	1.1279E-02	1.2498E-02	7.8773E-06	2.5376E-02	4.5220E-03	6.5569E-03	1.1212E-02	1.2194E-02
F_{16}	Mean	-1.0316E+00	-1.0316E+00	-1.0316E+00	-1.0316E+00	-1.0316E+00	-1.0316E+00	-1.0316E+00	-1.0304E+00	-9.5001E-01	-1.0213E+00
	Std	1.3924E-11	1.6650E-08	4.4409E-16	3.0520E-08	5.0139E-08	1.5803E-09	2.0477E-09	1.6732E-03	2.4485E-01	1.7032E-02
F_{17}	Mean	3.9790E-01	3.9789E-01	3.9789E-01	3.9789E-01	3.9789E-01	3.9789E-01	3.9789E-01	3.9885E-01	3.9789E-01	4.4809E-01
	Std	1.9471E-05	0.0675E-07	0.0000E+00	1.1738E-06	1.1336E-06	0.0000E+00	9.1969E-08	1.3513E-03	1.3746E-06	1.1747E-01
F_{18}	Mean	3.0000E+00	3.0000E+00	3.0000E+00	3.0000E+00	3.0000E+00	3.9000E+00	5.7000E+00	3.0000E+00	1.5672E+01	7.2958E+00
	Std	4.5594E-07	9.8650E-07	1.1665E-15	6.7184E-05	1.4382E-06	4.8466E+00	8.1000E+00	2.8203E-05	2.1799E+01	7.9626E+00
F_{19}	Mean	-3.8612E+00	-3.8628E+00	-3.8628E+00	-3.8598E+00	-3.8589E+00	-3.3449E+00	-3.8626E+00	-3.8516E+00	-3.7076E+00	-0.2323E+00
	Std	1.9702E-03	1.1593E-05	1.3323E-15	4.4800E-03	2.7441E-03	8.2955E-01	8.8539E-04	3.2373E-02	2.2722E-01	5.0990E-01
F_{20}	Mean	-3.1271E+00	-3.2578E+00	-3.2747E+00	-3.2727E+00	-2.7303E+00	-3.1122E+00	-3.2757E+00	-3.1569E+00	-3.2497E+00	-2.3278E+00
	Std	5.0973E-03	6.3850E-02	5.8399E-02	6.6943E-02	4.3592E-01	1.8591E-01	7.5459E-02	7.8861E-02	8.8987E-02	5.5817E-01
F_{21}	Mean	-1.0153E+01	-9.3663E+00	-8.9815E+00	-9.9023E+00	-1.9490E+00	-4.6848E+00	-5.4584E+00	-5.0458E+00	-9.0626E+00	-1.0136E+01
	Std	2.2244E-10	2.2472E+00	2.3982E+00	1.3406E+00	1.4743E+00	8.5352E-01	2.0391E+00	1.6104E+00	2.4873E+00	1.7467E-02
F_{22}	Mean	-1.0403E+01	-1.0139E+01	-1.0403E+01	-1.0224E+01	-2.5128E+00	-5.2505E+00	-5.8812E+00	-6.2381E+00	-7.9692E+00	-1.0383E+01
	Std	7.2852E-06	1.3729E+00	3.6260E-15	9.5376E-01	1.6017E+00	1.9977E+00	2.0509E+00	2.5930E+00	3.4380E+00	2.5042E-02
F_{23}	Mean	-1.0536E+01	-1.0277E+01	-1.0105E+01	-1.0535E+01	-2.7352E+00	-5.0657E+00	-4.9861E+00	-5.7580E+00	-8.2889E+00	-1.0527E+01
	Std	4.9283E-04	1.3903E+00	1.6130E+00	1.0204E-03	1.3818E+00	1.8974E+00	1.2902E+00	2.1694E+00	3.4710E+00	1.0145E-02

followed by ISSA. With improvements, the ISSA has made progress in fixed-dimension multimodal functions, but it has not performed very well.

Overall, the ISSA is very good at solving the unimodal test function compared with the original algorithm and other classical algorithms. This proves that our improvements have been effective the mining power of the algorithm. The ISSA also performs well in multimodal test functions, and it can be considered that our handling of the ability to explore and develop balanced algorithms is effective. However, in terms of fixed-dimension multimodal functions, although our improvement is better than the original algorithm, it does not perform very well. A fixed-dimension multimodal test function is a function that combines multiple simple functions. Different functions have different properties, such as multimodality, nonlinearity, noise, etc., to simulate the complexity of the real world. Therefore, there is still room for improvement in the face of complex problems.

4.2. Results comparisons using CEC2022

To have a more comprehensive understanding of the optimization ability and stability of the ISSA, we solve the ISSA with some improved algorithms using CEC2022, which are more complex and representative than the previous 23 standard test functions. There are a total of 12

single-objective test functions in the CEC2022, which are unimodal function (F_1), basic functions (F_2 - F_5), hybrid functions (F_6 - F_8) and composition functions (F_9 - F_{12}). F_1 is a new unimodal function obtained by displacement and rotation. F_2 - F_5 are new functions that are displaced, scaled, and rotated. F_6 - F_8 is the displacement of the variable, and then divided into multiple parts, different parts are calculated using different functions, and these functions are summed to obtain a new function. F_9 - F_{12} are variables that go through different displacements and rotations to obtain a number of new variables, and different functions are added to the new variables. Because of the transformation of the variables, the computation of F_1 - F_5 is more complex than that of the original function. F_6 - F_{12} , on the other hand, not only transforms variables, but also contains the characteristics of multiple functions in each function. This requires higher optimization capabilities of algorithms. The four improved algorithms are the Improved Fruit Fly Optimization Algorithm(WFOA) (Fan et al., 2020), Differential Evolution with Dynamic Self-adaptive (DESAP) (Teo, 2006), Adaptive Bat-inspired Algorithm(ABA) (Yang, 2010) and Adaptive Bacterial Foraging Optimization(ABFO) (Thieu, Minh, and Giang, 2019). We set the dimension of the CEC2022 test function to 20, each algorithm solved 30 times independently, 30 for each algorithm individual, and 200 iterations.

Table 7

The algorithm's ranking of the mean and standard deviation.

Index	Algorithm	ISSA	PSO	DE2	GWO	WOA	HHO	BES	SMA	SSA	AO
Mean	Average	2.83	7.22	4.96	5.91	6.7	3.7	5.43	4.87	5.91	5.78
	Ranking	1	10	4	7	9	2	5	3	7	6
Std	Average	2.52	7.3	5.74	6.09	5.91	4.48	3.83	4.7	5.7	6.91
	Ranking	1	10	6	8	7	3	2	4	5	9

Table 8

Experimental results of ISSA and other improved optimization algorithms on CEC2022.

Function	Index	WFOA	DESAP	ABA	ABFO	ISSA
F_1	Mean	3.82E+07	6.14E+04	7.82E+04	5.92E+04	4.89E+04
	Std	3.06E+07	1.92E+04	1.72E+04	1.42E+04	1.67E+04
F_2	Mean	5.51E+03	6.99E+03	8.79E+03	4.04E+03	3.90E+03
	Std	4.08E+02	2.28E+03	2.81E+03	8.26E+02	9.44E+02
F_3	Mean	6.02E+02	6.02E+02	6.03E+02	6.02E+02	6.02E+02
	Std	7.06E-02	4.01E-01	4.67E-01	1.90E-01	2.38E-01
F_4	Mean	1.16E+03	1.28E+03	1.35E+03	1.19E+03	1.14E+03
	Std	2.66E+01	5.95E+01	7.82E+01	7.48E+01	2.99E+01
F_5	Mean	9.22E+02	9.28E+02	9.35E+02	9.12E+02	9.10E+02
	Std	1.79E+00	8.73E+00	6.45E+00	3.85E+00	1.97E+00
F_6	Mean	8.75E+09	7.56E+09	1.06E+10	5.40E+09	8.74E+09
	Std	2.38E+09	2.59E+09	3.68E+09	2.08E+09	3.57E+09
F_7	Mean	4.78E+03	9.66E+03	1.18E+04	4.28E+03	8.89E+03
	Std	1.12E+03	2.92E+03	3.65E+03	8.31E+02	1.21E+03
F_8	Mean	2.81E+11	4.41E+13	3.71E+14	3.07E+12	8.53E+13
	Std	6.88E+11	9.12E+13	7.21E+14	5.15E+12	8.19E+13
F_9	Mean	1.56E+04	7.60E+03	8.76E+03	5.83E+03	1.41E+04
	Std	2.03E+03	2.26E+03	2.40E+03	7.96E+02	3.60E+03
F_{10}	Mean	5.25E+03	4.90E+03	5.77E+03	6.01E+03	2.96E+03
	Std	5.33E+02	6.77E+02	1.08E+03	8.03E+02	1.56E+02
F_{11}	Mean	3.25E+04	2.06E+04	2.57E+04	2.38E+04	1.10E+04
	Std	2.33E+03	8.33E+03	1.15E+04	7.18E+03	3.32E+03
F_{12}	Mean	4.69E+03	3.82E+03	3.83E+03	5.48E+03	3.50E+03
	Std	3.13E+02	2.90E+02	4.03E+02	5.25E+02	1.62E+02

Table 8 is the mean and standard deviation of the ISSA and other improved algorithms to solve the CEC2022. Fig. 10 is the convergence curve of the solution CEC2022 of the five algorithms, and the data is the iterative data that ranks 15th in the experimental solution accuracy of each algorithm. Fig. 11 is a sorted histogram of the CEC2022 solved by improved algorithms such as ISSA. Through Fig. 11, we can more intuitively understand the ranking of the solution CEC2022 of the five algorithms. Combined with the Table 8 and Fig. 11 analysis, ISSA ranked first on F_1 , F_2 , F_5 , F_{10} , F_{11} and F_{12} , followed by ABFO ranked first on F_3 , F_4 , F_6 , F_7 and F_9 , and WFO ranked first on F_8 . Looking at the top three algorithms, the ISSA ranks in the top three on 10 test functions, while ABFO is equally competitive with the ISSA. Combined with the first and top three cases, the solution accuracy and stability of ISSA and ABFO are generally similar. But the ISSA is slightly better than the ABFO. Fig. 10 analyzes the convergence speed of the five algorithms, and on F_1 and F_2 , the ISSA converges quickly and with high accuracy. The solution accuracy of ABFO is close to that of ISSA, but the convergence speed is lower than that of ISSA. On F_3 , F_4 , F_{10} , and F_{11} , the ISSA has high solution accuracy, but the convergence speed is not fast.

On the unimodal test function F_1 and the basic function F_2 - F_5 , the ISSA ranks first on three functions and second on two functions. It can be seen that the ISSA still performs well in solving more complex unimodal and multimodal problems. Hybrid functions and complex functions F_6 - F_{12} are test functions that are composed of multiple basic functions that have been processed and combined. When solving these functions, the ISSA performed best on the three test functions and ranked fourth on the two test functions compared to other improved algorithms. It can be seen that the ISSA has a certain competitiveness in the face of complex problems, but there is still room for improvement.

Finally, a brief explanation of the computational complexity of the algorithm is given. In our experiments, the population of the algorithm is N , the maximum number of iterations is $T_{\max}$, the dimension of the problem is dim , and the fitness function of a single individual solution

is $O(fit)$, the number of danger perceivers of the original algorithm is set to $0.15N$, and the number of danger perceivers of the ISSA is ϖN , where $\varpi \in [0.1, 0.2]$. Therefore, the complexity of the original SSA algorithm is $O(1.15 * N * T_{\max} * dim * fit)$, and the complexity of ISSA is $O(\varpi * N * T_{\max} * dim * fit)$.

5. ISSA based UAV path planning

In Section 4, it is verified that ISSA has some advantages in solving optimization problems by comparing ISSA with other classical or novel meta-heuristic algorithms, including the original SSA, in 23 standard test functions. Similar to the standard test function, the UAV path planning problem also has the characteristics of nonlinearity, discretization, and high dimensionality (Hu et al., 2023). Therefore, this paper uses ISSA to deal with UAV path planning problems for 2D maps containing circular and polygonal obstacles, and 3D maps containing mountain peaks and cylindrical obstacles. Demonstrate its ability to solve complex optimization problems by analyzing the paths generated by the ISSA. The mathematical objective function model of path planning can be described in the following form:

$$F(x) = f(x) + \lambda \sum_{j=1}^p g_j(x) \delta(g_j(x)), j = 1, 2, \dots, p \quad (14)$$

$$\delta(g_j(x)) = \begin{cases} 1, & g_j(x) > 0 \\ 0, & g_j(x) \leq 0 \end{cases} \quad (15)$$

where x represents a complete path, $F(x)$ is the objective function, $f(x)$ is the original fitness function, $g_j(x)$ is the constraint function, λ is the penalty parameter of the constraint, $\delta(g_j(x))$ is the constraint condition of the constraint function, and p is the number of constraints. In the 2D map, constraints include threat constraint and UAV maximum turn constraint. In the 3D map, the maximum climb angle constraint of the UAV is also included.

To verify the superiority and robustness of the ISSA algorithm, we compare it with the PSO, GWO, WOA, and SSA through four

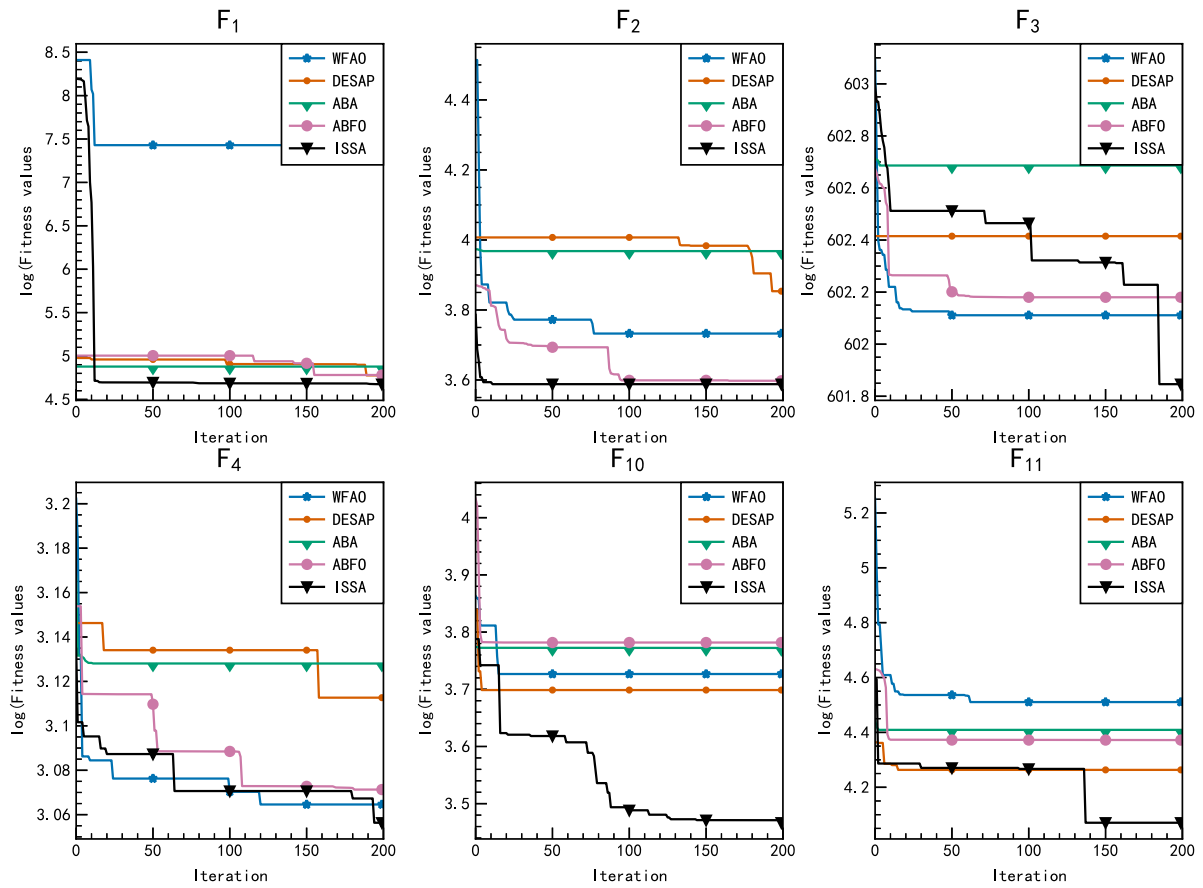


Fig. 10. Convergence curves of ISSA and other improved optimization algorithms on CEC2022.

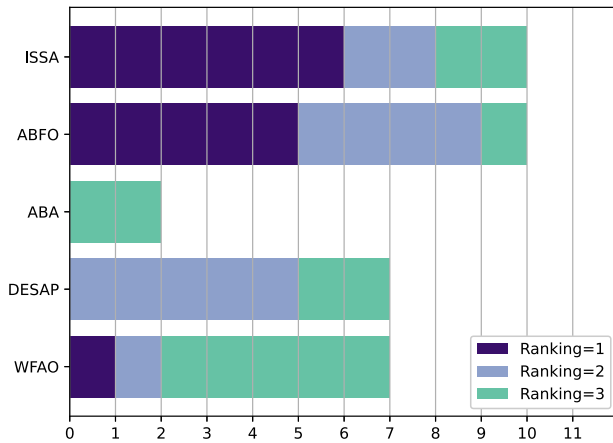


Fig. 11. Histogram of the ordering of algorithms such as ISSA on CEC2022.

test instances designed in 2D and 3D environments. The parameters of these algorithms are consistent with those of the experiments in Section 4. For a fair comparison, these algorithms have a maximum of 200 iterations and 30 individuals, all algorithms are run independently 30 times repeatedly, and the resulting paths and statistics are used to compare their performance.

5.1. Results and analysis of 2D maps

In the 2D map experiment, we designed three cases to demonstrate the performance of each algorithm by setting different threat numbers,

threat shapes, and placement complexities. The first case has a total of 9 circular threats, the second case contains 20 circular threats, and the third case consists of 24 squares and rectangles of different sizes forming 14 polygons. Table 9 gives detailed information about the coordinates and radii of the circular threats for the first and second cases. In addition, Table 9 gives the coordinates of the quadrilaterals as well as length and width information for the third case. The start and end points of all three cases are set to (0,0) and (1000,1000), respectively.

In the first case, the number of threats is 9 and the results are shown in Fig. 12. It shows the difference in the statistical indicators and the difference in the path of the 5 algorithms. It is easy to see from the graph that all algorithms have found a safe path. The ISSA generates fewer curves and smoother paths. While WOA and SSA paths also get feasible solutions but are more curved than the previous three algorithms, and these two algorithms cost more than the other three algorithms.

In the second case, 20 circle-centered threat sources of different sizes are included. As can be seen in Fig. 13, all five algorithms find feasible solutions. The four paths, except for WOA, have small turns and smooth paths. Among the five algorithms, the best result is the ISSA algorithm, and the best stability is SSA, but the mean value of 30 simulations for SSA is not good, only better than WOA.

In the third case, a total of 14 irregular obstacles are included. As can be seen in Fig. 14, the paths of WOA and SSA are relatively less smooth and have more turns than the other algorithms. In contrast, the paths of GWO and PSO almost overlap with fewer turns and smoother paths. From the bar chart in Fig. 14 it can be seen that PSO has good stability and stable results, the SSA is also stable but has a large mean value. WOA and ISSA are less stable and from Table 11 it can be seen

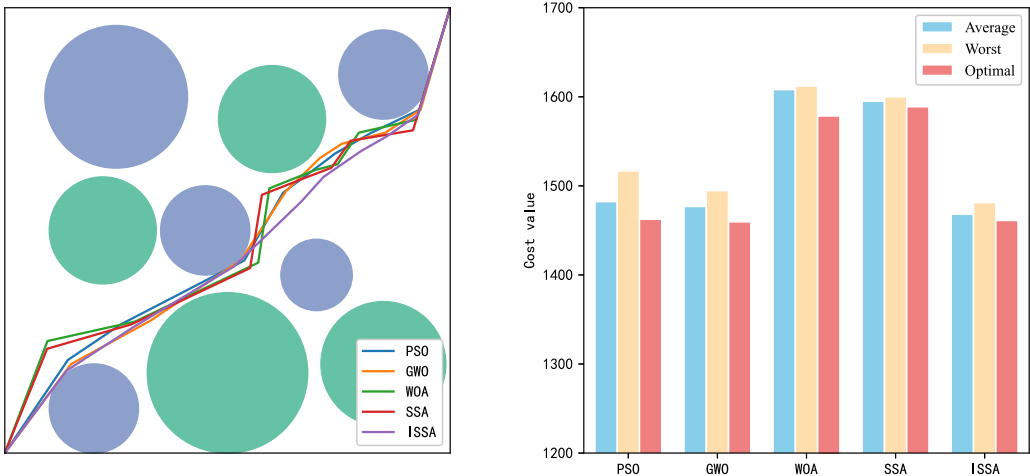


Fig. 12. Results of five algorithms for case 1.

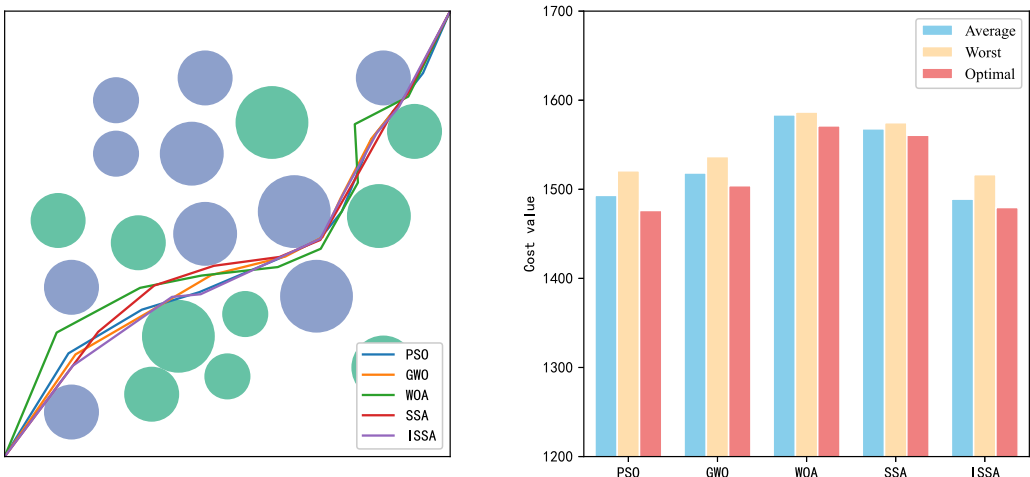


Fig. 13. Results of five algorithms for case 2.

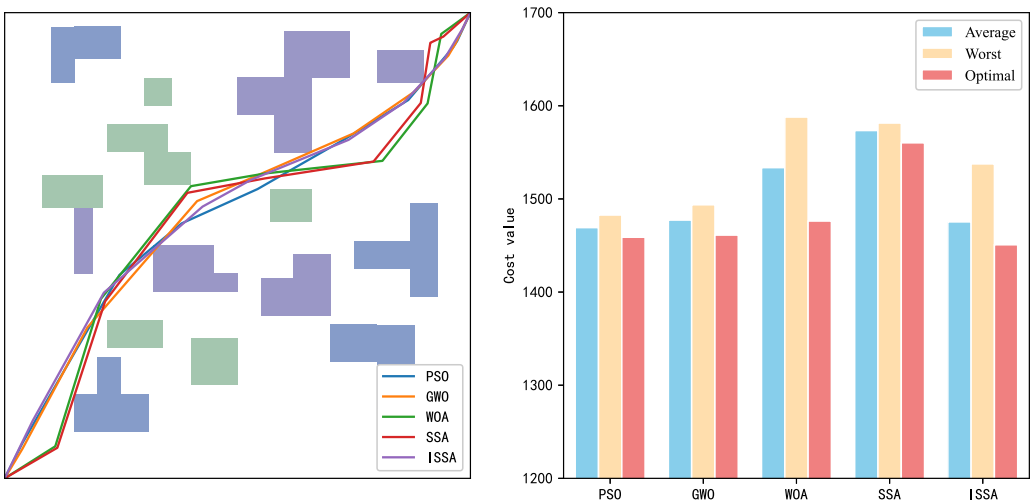


Fig. 14. Results of five algorithms for case 3.

Table 9
Detailed data for case 3.

Case 1 (x, y, r)	Case 2 (x, y, r)	Case 3 (x, y, w, h)			
(200, 100,100)	(150, 100,60)	(150, 380,60)	(300, 800,60, 60)	(580, 700,80, 80)	(800, 850,100, 70)
(700, 400,80)	(700, 360,80)	(650, 550,80)	(400, 200,100, 100)	(600, 860,140, 100)	(320, 400,130, 100)
(250, 800,160)	(250, 800,50)	(250, 680,50)	(570, 550,90, 70)	(550, 350,150, 80)	(450, 400,50, 40)
(450, 500,100)	(450, 500,70)	(420, 680,70)	(220, 280,120, 60)	(620, 430,80, 50)	(500, 780,160, 80)
(850, 850,100)	(850, 850,60)	(450, 850,60)	(300, 630,100, 70)	(150, 100,160, 80)	(700, 250,100, 80)
(220, 500,120)	(120, 530,60)	(330, 140,60)	(220, 700,130, 60)	(200, 180,50, 80)	(800, 150,80, 180)
(600, 750,120)	(600, 750,80)	(390, 270,80)	(80, 580,130, 70)	(100, 850,50, 120)	(750, 450,120, 60)
(500, 180,180)	(500, 180,50)	(540, 320,50)	(150, 440,40, 140)	(150, 900,100, 70)	(870, 390,60, 200)
(850, 200,140)	(850, 200,70)	(840, 540,70)			
	(300, 480,60)	(920, 730,60)			

Table 10
Detailed data for case 4.

Peaks			Cylinders	
(x, y)	(a, b)	h	(x, y)	r
(25,25)	(11,16)	2	(10,20)	5
(16,70)	(10,8)	1.3	(50,45)	7
(50,20)	(12,10)	2	(70,80)	8
(45,78)	(14,7)	2	(88,15)	7
(67,59)	(9,6)	2		
(78,80)	(10,8)	1.5		

Table 11
Experimental results for 4 cases.

Algorithm	Index	Case1	Case2	Case3	Case4
PSO	Mean	1482.28	1493.20	1469.18	156.63 [t]
	Max	1516.78	1520.80	1482.70	286.80
	Std	15.63	12.24	6.04	34.41
	Min	1462.51	1476.23	1458.87	142.44
GWO	Mean	1608.05	1583.54	1533.58	151.40
	Max	1611.99	1586.70	1587.87	154.07
	Std	7.74	4.00	47.66	2.83
	Min	1578.46	1571.21	1476.24	148.10
WOA	Mean	1595.05	1567.89	1573.41	162.19
	Max	1599.96	1574.74	1581.46	209.23
	Std	3.18	4.20	5.55	12.63
	Min	1588.83	1560.66	1560.23	155.04
SSA	Mean	1476.89	1518.31	1477.28	148.16
	Max	1494.66	1536.71	1493.71	207.17
	Std	8.92	8.40	8.17	11.00
	Min	1459.61	1504.03	1461.15	142.32
ISSA	Mean	1468.17	1488.96	1475.33	147.98
	Max	1481.32	1516.46	1537.60	151.54
	Std	4.46	7.09	22.04	1.28
	Min	1461.16	1479.55	1450.77	146.38

that in the third instance, ISSA has the second highest mean value and the stability is only better than that of WOA. PSO is the best performing algorithm in case 3.

5.2. Results and analysis of 3D map

In the actual flight of UAV, many irregular 3D obstacles will be encountered. Therefore, in the fourth case, we use the peak function to simulate hills and the cylindrical obstacle model to simulate threat areas. The mathematical expression of the hill peak function is:

$$h(x, y) = \sum_i h_i \exp\left[-\frac{(x - x_{mi})^2}{a_i^2} - \frac{(y - y_{mi})^2}{b_i^2}\right] + h_m$$

where h_m and h_i denote the heights of the base terrain and the i^{th} peak, respectively, h_m is set to 0, (x_{mi}, y_{mi}) denotes the center position of the i^{th} peak, a_i and b_i denote the slopes along the x -axis and y -axis of the i^{th} peak, respectively. Table 10 shows the detailed data of the peaks and the coordinates, radius and height of the cylindrical barriers. Fig. 15 shows the terrain simulation of the 3D environment.

In the fourth case, the 3D environment contains six peaks and four cylindrical threat zones. As can be seen in Fig. 16 each algorithm finds

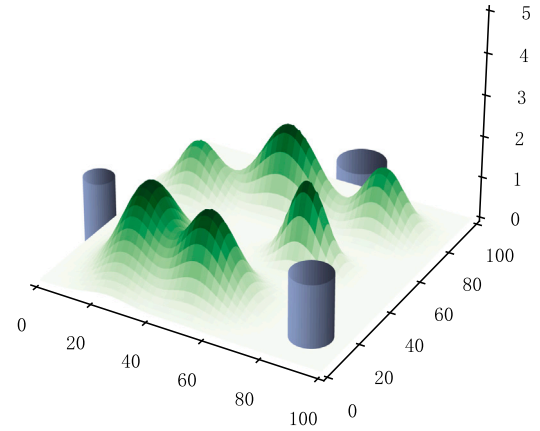


Fig. 15. 3D terrain simulation map.

a feasible path. The smoothest path is ISSA, which traverses through two peaks and passes through the cylindrical threat zones at a higher flight altitude than the threat zones. As can be seen from the bars in Fig. 16, the better results are WOA and ISSA, which are two algorithms with low mean values and good stability. While the mean values of the four cases do not differ much, the other three algorithms are poorly stabilized, especially the PSO algorithm, where the maximum value is much higher than the minimum value, showing a relatively unstable characteristic.

6. Conclusions and future works

In this paper, an improved SSA based on uniform design theory is proposed and successfully applied to the UAV path planning problem. In the UAV flight environment modeling, four scenarios of circles, irregularly shaped obstacles, columns, and mountains are constructed. The proposed ISSA introduces four new improvement strategies in SSA. Utilizing good lattice point designs and TA to generate the initial population enhances the population diversity, expands the initial solution space, and improves the convergence speed of SSA. Macroscopically adjusting the position update strategy of the producer not only avoids the population convergence to the origin to widen the search area but also balances the exploration and exploitation ability of the algorithm and effectively prevents the algorithm from falling into the local optimum prematurely. Then a dynamic population uniformity affiliation function is introduced to adjust the number of danger perceivers by adjusting the affiliation function. Finally, a new boundary update strategy is proposed to adjust the position of individuals exceeding the boundary based on population uniformity. We use 23 standard test functions to verify the performance of ISSA and compare it with 9 classical and novel meta-heuristic algorithms. In addition, we used CEC2022 to validate the performance of the ISSA and compare it with four improved algorithms. Experiments show that ISSA performs well in solving unimodal and multimodal problems. This means that our improvements

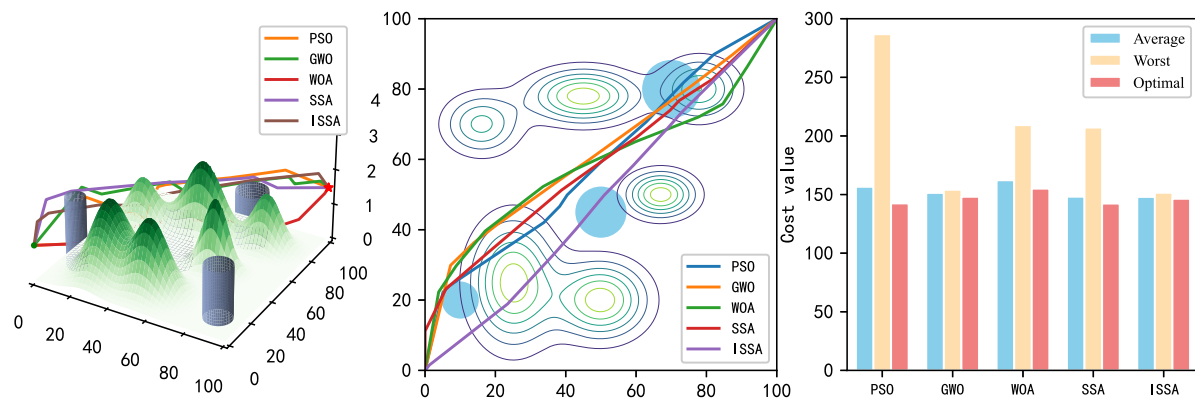


Fig. 16. Results of five algorithms for case 4.

improve the exploitation ability of the algorithm and the ability of the algorithm to balance exploration and exploitation. Compared with the original SSA, the algorithm's ability to solve complex problems has improved, but it can continue to improve. It succeeds in obtaining good flight paths in four different scenarios of path planning and excels in stability. In our paper, we focus on improving the algorithm on how to prevent the algorithm from converging prematurely and falling into local optimums. Therefore, our focus has improved the finder formula and adjusted the number of scouts, without modifying the followers. In addition, we did not consider combining with other meta-heuristic algorithms to improve the SSA and improve the solution accuracy of the algorithm by using the advantages of other algorithms.

Consider the possibility of fusing the improved algorithm with other optimization algorithms, and make full use of the advantages of different algorithms to enhance the overall searchability. This will help solve more complex optimization problems and improve the versatility of the algorithm. Because of the excellent performance of ISSA in dealing with single-objective problems and the wide application of multi-objective optimization in practical problems, it is worth exploring and studying the application of existing improved strategies to multi-objective scenarios. In addition, the potential of the ISSA in other applications can be further explored. For example, in the field of hyperparameter optimization of robots.

CRedit authorship contribution statement

Lianyu Cheng: Visualization, Writing – original draft. **Guang Ling:** Methodology, Supervision, Writing – review & editing. **Feng Liu:** Investigation, Resources. **Ming-Feng Ge:** Conceptualization, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The relevant data are all in the table of the article.

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